

Project 5: Full Box–Jenkins ARMA study with forecasting

Identification · Estimation · Diagnostics · Forecast

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Context

This project consolidates all of Chapters 3–5: stationarisation, ACF/PACF identification, ARMA estimation, residual diagnostics and forecasting. You will produce a complete Box–Jenkins study on the Souvenir Sales data, on the differenced log-scale, where an ARMA model is appropriate.

The deliverable serves two purposes: it will be re-used in Chapter 6 (ARCH/GARCH) where you will check the residuals for volatility clustering.

Deliverables

- Reproducible `project5_<name>.py` with seed 42.
- Report PDF (8–10 pages) including all figures, all numerical outputs, and a final back-transformed forecast with 95% CI.
- Optional 10-min oral defence.

Exercise 1 – Stationarisation

- (1) Load `data/SouvenirSales.csv`. Compute $y_t = \log(\text{sales}_t)$.
- (2) Apply the seasonal log-difference $w_t = (1 - L)(1 - L^{12})y_t$. Verify by ADF/KPSS that w_t is stationary.
- (3) From now on, model w_t as an *ARMA* on the stationary scale. Plot w_t , its sample ACF and PACF up to lag 36.

Exercise 2 – Identification

- (1) Read the ACF/PACF of w_t at the regular lags (1–11): which (small) AR or MA orders look plausible?
- (2) Read at the seasonal lags (12, 24, 36): does the spike at lag 12 persist or disappear after one differencing?
- (3) Propose three candidate orders, e.g. (0, 1), (1, 0), (1, 1) on the regular part (and you may add a seasonal component if needed; see Chapter 4).

Exercise 3 – Estimation and selection

- (1) Fit each candidate by MLE (`statsmodels.tsa.arima.ARIMA`). For each, report:
 - the estimated coefficients with standard errors;
 - the log-likelihood, AIC, BIC, $\hat{\sigma}^2$;
 - the Ljung–Box p -value at $m = 12$ and $m = 24$.

- (2) Choose the best model on the (AIC, BIC, Ljung–Box) triplet. Justify in 5 lines.
 (3) Plot the model residuals: time plot, sample ACF, QQ-plot vs $\mathcal{N}(0, 1)$. Are they convincingly white noise?

Exercise 4 – Forecasting on the stationary scale

- (1) On the chosen ARMA, produce a 12-step forecast of $w_{T+1}, \dots, w_{T+12}$ with 95% CI.
 (2) Check that the forecast returns to the unconditional mean $\mathbb{E}[w] \approx 0$ as h grows.
 (3) Plot the in-sample fit and the forecast with the CI band.

Exercise 5 – Back to the original scale

- (1) Recall $w_t = (1 - L)(1 - L^{12})y_t$. Inverting one step at a time, derive the recursion that takes $\hat{w}_{T+h}$ back to $\hat{y}_{T+h}$ (telescope sum on the seasonal and the regular difference).
 (2) Implement the back-recursion in Python. Plot y_t together with $\hat{y}_{T+h}$ and the back-transformed CI.
 (3) Take the exponential to recover sales:

$$\widehat{\text{Sales}}_{T+h} = \exp(\hat{y}_{T+h} + \hat{\sigma}_h^2/2)$$

(conditional mean) or $\exp(\hat{y}_{T+h})$ (median). Report the predicted sales for the 12 months following December 2001.

Exercise 6 – Bonus: hold-out validation

- (1) Split: train = Jan 1995 – Dec 2000 (72 obs), test = Jan – Dec 2001 (12 obs).
 (2) Re-fit the chosen ARMA on the training set (with the same differencing recipe), forecast the test horizon, back-transform.
 (3) Report MAE, RMSE, MAPE on the original scale. Discuss: does the model under- or over-estimate the December peak?

Marking grid (out of 100)

Criterion	Points
Stationarisation + ADF/KPSS verification	10
ACF/PACF identification of three candidates	15
MLE estimation + AIC/BIC selection	20
Residual diagnostics (time, ACF, QQ, LB)	15
Forecast on the stationary scale + CI	10
Back-transform to the original scale	10
Bonus: hold-out validation (RMSE/MAPE)	10
Reproducibility + report quality	10
Total	100

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